MIL-P-10590B(MU)
24 October 1966
SUPERSEDING
MIL-P-10590A(MU)
17 January 1963

MILITARY SPECIFICATION
PROJECTILE, ILLUMINATING, 81MM, M301A2
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, Illuminating, 81MM, M301A2, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-10025	- Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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DRAWINGS

Munitions Command

D10523459	- Projectile, Illuminating, 81MM, M301A2, Metal Parts Assembly
IEL-10523459	- Index to Inspection Equipment List for Projectile, Illuminating, 81MM, M301A2, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for submission of first article sample by the contractor shall be as specified in the contract.

3.3 Heat treatment or stress relief.- Heat treatment or stress relief shall be uniform throughout each lot of components and shall be so conducted as to produce the mechanical properties specified on the applicable drawing.

3.3.1 Hot forged heat treat method.- Prior to finish machining, projectile components manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, followed by an oil quench. After quenching, the body forgings shall be tempered without delay at a temperature suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700° F., after which cooling may be accelerated by water spray.

3.3.2 Cold shaping and hot cup - cold draw methods.- Cold working shall be so accomplished as to produce in the projectile components the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700° F., minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified on the applicable drawing. The temperature and time duration of a process anneal prior to final forming operation shall be established by the ability to meet the required mechanical properties when formed by an approved sequence of operations.

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3.4 Mechanical properties.- All components shall meet the mechanical properties requirements as specified on the applicable drawings.

3.5 Metal defects.- All components shall be free from cracks, seams, splits, cold shuts, inclusions, porosity or similar defects.

3.6 Workmanship.- The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.6.1 Surface finish.- The requirements for surface are as detailed on the applicable drawings. The cavity shall be free of burrs, scale, fins, deep seams, pits, or other surface discontinuities.

3.6.2 Protective coating.- Protective coating shall be in accordance with the applicable drawing and the specifications listed thereon.

3.6.3 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of Specification MIL-A-2550.

3.6.4 Burr.- No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.6.5 Foreign matter.- No part or assembly shall contain dirt, grease, rust, chips, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with specification MIL-A-2550 except as specified herein.

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4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of the following:

5 metal parts assemblies, complete (painted)
5 metal parts assemblies, complete (unpainted)
5 sets component parts, unassembled (unpainted)
2 Cones, tail, prior to machining

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Heat treatment lots.- Heat treatment lots shall contain component forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Stress relieved lots.- Stress relieved lots shall contain cold worked components from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.4 Miscellaneous lots.- Small lots (such as components manufactured from abnormally small heats of steel that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.3.1.2 or 4.3.1.3 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.5 Projectile lots.- When it has been determined that components of a heat-treated stress relieved or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawing, they may be re-grouped for assembly into projectile lots, to which will be assigned a projectile lot number. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or workweek.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual inspection for such defects as burr and foreign matter.

4.3.2.1 Classification of defects.

4.3.2.1.1 Cone, tail (prior to paint), Drawing C10523455 covering a detail of Drawing D10523459.

Defect	Method of Inspection
CRITICAL	
None defined	

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4.3.2.1.1 Cont'd

Defect		Method of Inspection
MAJOR		
101.	Concentricity of pitch diameter of fin thread with forward outside diameter, including perpendicularity of body seat and rear face	Gage
102.	Parallelism of front face with body seat	Gage
103.	Perpendicularity of body seat with outside forward diameter	Gage
104.	Diameter outside, forward	Gage
105.	Diameter inside at front	Gage
106.	Diameter of shoulder	Gage
107.	Diameter of hole through base	Gage
108.	Length of forward diameter, outside	Gage
109.	Metal defective	Visual
MINOR		
201.	Diameter at rear of shoulder	Gage
202.	Diameter, small outside at rear	Gage
203.	Depth of undercut forward of body seat, max. (if applicable)	Gage
204.	Length from rear face to start of taper	Gage
205.	Length from rear face to front of taper	Gage
206.	Length from rear face to shoulder	Gage
207.	Length total	Gage
208.	Thickness through base to rear face	Gage
209.	Thickness of wall	Gage
210.	Width of undercut forward of body seat, max. (if applicable)	Gage
211.	Surface finish, improper	Visual (see 6.1)
212.	Radii or chamfers missing or incorrect	Visual (see 6.1)
213.	Staking notches missing or incorrect	Visual (see 6.1)
214.	Burr	Visual

4.3.2.1.2 Cone, tail (after paint) Drawing C10523455 covering a detail of Drawing D10523459.

CRITICAL

None defined

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4.3.2.1.2 Cont'd

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of fin thread, max.	Gage
102.	Minor diameter of fin thread, max.	Gage
MINOR		
201.	Pitch diameter of fin thread, min.	Gage
202.	Diameter of fin thread, undercut max. (if applicable)	Gage
203.	Diameter of fin thread counterbore	Gage
204.	Depth of fin thread counterbore	Gage
205.	Depth from rear face to bottom of fin thread cavity	Gage
206.	Length of effective fin thread, min.	Gage
207.	Silicone compound missing, threads	Visual
208.	Silicone compound missing forward outside diameter	Visual
209.	Paint on forward outside diameter	Visual
210.	Forward outside diameter damaged	Visual
211.	Threads damaged or not full	Visual
212.	Foreign matter	Visual

4.3.2.1.3 Pin Adapter, Drawing B10523454 covering a detail of Drawing D10523459.

CRITICAL		
None defined		
MAJOR		
101.	Diameter	Gage
MINOR		
201.	Length total	Gage
202.	Chamfer missing or incorrect	Visual (see 6.1)
203.	Surface finish, improper	Visual (see 6.1)
204.	Burr	Visual
205.	Foreign matter	Visual

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4.3.2.1.4 Adapter, Drawing C10523456 covering a detail of Drawing D10523459.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter at rear of shoulder	Gage
102. Diameter of shoulder	Gage
103. Metal defective	Visual
MINOR	
201. Diameter at front	Gage
202. Length from front face to shoulder	Gage
203. Length total	Gage
204. Staking notches missing or incorrect	Visual (see 6.1)
205. Radii or chamfers missing or incorrect	Visual (see 6.1)
206. Surface finish, improper	Visual (see 6.1)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.1.5 Tube, Body, Drawing C10523457 covering a detail of Drawing D10523459.

CRITICAL	
None defined	
MAJOR	
101. Front face not parallel with rear face	Gage
102. Diameter of adapter cavity	Gage
103. Diameter of bourrelet, min.	Gage
104. Diameter of charge cavity	Gage
105. Diameter of cone cavity	Gage
106. Diameter of body outside	Gage
107. Thickness of wall, min.	Gage
108. Metal defective	Visual

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4.3.2.1.5 Cont'd

Defect	Method of Inspection
MINOR	
201. Depth of adapter cavity	Gage
202. Depth of cone cavity	Gage
203. Length of bourrelet	Gage
204. Length, total	Gage
205. Surface finish, improper	Visual (see 6.1)
206. Radii or chamfers missing or incorrect	Visual (see 6.1)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.1.6 Metal Parts Assembly (prior to painting) Drawing D10523459.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of pitch diameter of threads with cone cavity including perpendicularity of front and rear faces	Gage
102. Concentricity of bourrelet with cone cavity	Gage
103. Concentricity of body with cone cavity	Gage
104. Length total, max.	Gage
105. Pin(s) missing	Visual
MINOR	
201. Diameter pin hole	Gage
202. Depth of pin hole	Gage
203. Length from front face to pin hole	Gage
204. Location of pins circumferential	Visual (see 6.1)
205. Radii or chamfer missing or incorrect	Visual (see 6.1)
206. Pin(s) not peened (if applicable)	Visual
207. Marking missing, incorrect or illegible	Visual
208. Burr	Visual
209. Foreign matter	Visual

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4.3.2.1.7 Metal Parts Assembly (after painting) Drawing D10523459.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Weight	Approved scale
102.	Pitch diameter of thread, max.	Gage
103.	Minor diameter of thread, max.	Gage
104.	Diameter of bourrelet, max.	Gage
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Interior paint, inadequate	Visual
203.	Exterior paint, inadequate or damaged	Visual
204.	Paint on rear inside diameter	Visual
205.	Silicone compound missing, rear inside diameter	Visual
206.	Silicone compound missing, threads	Visual

4.3.2.1.8 Packed carton prior to sealing.

CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Support missing or damaged	Visual
202.	Separator missing or damaged	Visual
203.	Improper assembly	Visual
204.	Incorrect number in carton	Visual

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4.3.2.1.9 Packed carton after sealing.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton not completely sealed	Visual
203. Contents loose	Manual/Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Tension test.

4.3.3.1.1.1 Body tube.- After final heat treatment or stress relief, two body tubes shall be selected from each heat treated or stress relieved lot of body tubes and subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable component drawing. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.1.1.2 Cone, tail.- After final heat treatment or stress relief, two tail cones shall be selected from each heat treated or stress relieved lot of tail cones and subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable component drawing. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.1.1.3 Adapter.- The sample size and acceptance criteria shall be as specified in the applicable material specification, except that the mechanical properties specified on the applicable component drawing shall apply to the finished components. Failure to comply with the requirements of the applicable drawing and specifications shall be cause for rejection of the lot.

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4.3.3.1.1.4 Miscellaneous lots.- Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.3.3.1.1.1 or 4.3.3.1.1.2, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Double the number of samples shall be selected for miscellaneous lots.

4.3.3.2 Protective coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.4 Inspection equipment.- Index of Inspection Equipment List Number 10523459 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable IEL. Reference is made to the inspection equipment list index for definition of coding format.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Body tube.- From each sample selected in accordance with 4.3.3.1.1.1 or 4.3.3.1.1.4, two tension test specimens conforming to the Type T-1 prescribed by Federal Test Methods Standard No. 151 shall be taken longitudinally 180 degrees apart. Yield strength shall be determined by either the offset method or the extension under load method as prescribed by Federal Test Method Standard No. 151. The limiting set shall be 0.1 percent (0.001 inches per inch of gage length).

4.4.1.2 Cone, tail.- From each sample selected in accordance with 4.3.3.1.1.2 or 4.3.3.1.1.4, two tension test specimens of the largest obtainable R type specimen conforming to the prescribed Federal Test Method Standard No. 151 shall be taken longitudinally from the straight portion of the component, prior to machining, 180 degrees apart. Yield strength shall be determined by either the offset method or the extension under load method as prescribed by Federal Test Method Standard No. 151. The limiting set shall be 0.1 percent (0.001 inches per inch of gage length).

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4.4.2 Protective coating.- The procedures for testing the cleaning of components and assemblies and the application of protective coating shall be in accordance with the applicable specifications listed on the assembly drawing. The salt spray test shall be performed utilizing the equipment prescribed by Federal Test Method Standard No. 151. A 20 percent sodium chloride solution shall be used.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. - Not applicable.

5.1.2 Level B. - Not applicable.

5.1.3 Level C. - Metal parts assemblies shall be packed in accordance with Specification MIL-P-10025.

5.2 Labeling and marking.- Labeling and marking shall be in accordance with Specification MIL-P-10025.

5.3 Data card.- Data card information shall be as specified in Specification MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Heat treatment is understood to be any thermal process which brings out a modification in the mechanical or metallurgical properties.

6.3 Ordering data.- Procurement documents should specify the title, number and the date of this specification.

Custodian:

Army - MU

Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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